

SHAURYA SHARTHAK

Python Software Engineer

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PROFESSIONAL SUMMARY

Python software engineer with strong open-source contributions (234 GitHub stars) and production experience in **backend systems**, **data pipelines**, and **API development**. Built distributed training frameworks, automated retrieval systems, and CLI tools. Experienced in **system design**, **CI/CD**, and **cloud infrastructure (AWS)**. Founded Tinycompany-AI managing 20+ repositories. Published first-author research requiring rigorous software engineering for reproducible experiments.

EXPERIENCE

AI Engineer (Intern)

08/2024 – 01/2026

ResoluteAI Software Inc.

Bangalore, India

- **Backend Pipeline Engineering:** Architected hybrid retrieval service (TF-IDF + BM25 + Neo4j + Redis + vector stores) with modular Python components, improving hit-rate by **45%**.
- **Performance Optimization:** Refactored inference pipelines for **30% latency reduction** and **Rs.25.8 Lakhs/year** cost savings through profiling and bottleneck elimination.
- **Automation System:** Developed Pandas agent-based ticket automation with REST API integration, gaining **30% ops efficiency**.
- **Engineering Practices:** Agile sprints, Git feature branching, Postman API testing, Jenkins CI/CD pipelines.

TECHNICAL SKILLS

Languages: Python (Expert), SQL, C++, Bash, LaTeX

Python Ecosystem: PyTorch, Flask, Pandas, NumPy, Asyncio, Requests, BeautifulSoup, Selenium, Click/Argparse

Backend & APIs: REST APIs, Postman, LangChain, HuggingFace Transformers/Datasets

Databases: PostgreSQL, Redis (KV), Neo4j (Graph), ChromaDB/Pinecone (Vector)

DevOps & Cloud: Git/GitHub, Docker, Jenkins CI/CD, AWS (EC2/S3), Linux/Bash scripting

Testing & Tools: Unit Testing, Profiling, Jupyter, MinHash-LSH, Web Scraping

KEY PROJECTS

PyTorch/XLA TPU Training Framework | 234 stars, 27 forks

2024

- Engineered end-to-end distributed training framework for **TPU v2-8** in Python; resolved XLA graph-compilation vs. eager-execution incompatibilities for Llama and Mistral architectures.
- Built **Optimized-Attention-Torch-XLA** — drop-in Flash/Splash Attention module compatible with XLA backend; custom trainer with **GSPMD model parallelism** reducing memory by **35%**.
- Maintained 3 interconnected repositories (TPU-Alignment, XLA-Trainer, Optimized-Attention) with clean APIs and documentation.

Open-Source CLI & Automation Tools | GitHub

2022 – 2025

- Built **AnimeDownloader** (5 stars, 6 forks) — Python batch downloader with web scraping and session management.
- Developed **Spotify-dl** (playlist downloader), **IRCTC-Reservation-Finder** (real-time seat scraper), **Microsoft-Rewards-Automator** — all Python CLI tools with modular architecture.
- Created **MinHash-LSH-DeDup** — batched deduplication library for large-scale datasets using locality-sensitive hashing.

TokenAdapt & SuperTokenizer — Research Software | 14 stars

2025

- Implemented **TokenAdapt** framework in Python — model-agnostic tokenizer transplantation with hybrid embedding initialization; reproducible benchmarking pipeline.
- Built **SuperTokenizer** (13 stars) — probabilistic multi-word pre-tokenization library with Gaussian chunk-length distribution.

Cybersecurity & Security Research

2024

- Developed **Mouse-OS** — AI-generated RAT for security research in Python.
- Performed penetration testing for IDORs, BOLA and Broken Access Control in **3+ MNC web apps**; secured systems handling **2M+ transactions**.

PUBLICATIONS

Tokenizer Flexibility in LLMs via Heuristic Adaptation and Supertoken Learning

arXiv:2505.09738 | 2025

- **Lead Author | 4 citations** — Designed and implemented reproducible experiment pipeline with **2×** perplexity improvement.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University

2022 – 2026

B.Tech (Hons.) Computer Science (AI/ML)

Bhilai, India

Jawahar Navodaya Vidyalaya

2018 – 2021

Senior Secondary (GPA: 3.86/4.0) — Secondary (GPA: 3.6/4.0)

Kurukshetra, India